INTRODUCTION TO DBMS

DATABASE MANAGEMENT SYSTEM

Contents

1. What is Data?

2. What is a Database?

3. What is a DBMS?

4. Why Do We Need a DBMS?

5. File System vs DBMS

6. Main Components of a DBMS

7. Types of Database Models

8. SQL: The Language of Databases

9.Advantages and Limitations

10.Conclusion



1. What is Data?

Data refers to raw facts, figures, or details that are collected for reference or analysis. Data may include numbers, names, dates, symbols, images, or text. On its own, data may not provide a complete meaning, but when it is processed and organised, it becomes useful information.

Examples of data:

- Student name: Rahul
- Roll number: 101
- Marks: 85
- Age: 20
- Course: MCA

2. What is a Database?

A database is an organised collection of related data that is stored electronically so that it can be easily accessed, managed, updated and retrieved.

A database helps individuals and organisations maintain large amounts of information in a systematic manner. Instead of storing information in separate registers or files, a database allows users to keep everything organised in one place.

Example: A college database stores information about students, teachers, courses, attendance, fees and examination results.

Example of a Student Database:

STUDENT ID	STUDENT NAME	COURSE	MARKS
101	Rohini	MCA	90
102	Priya	MCA	97
103	Vansh	MCA	20
104	Aman	MCA	50
105	Pranay	MCA	10

Characteristics of a Database:

1. Stores related information in an organised way.
2. Allows easy searching and retrieval.
3. Supports adding, deleting and updating records.
4. Helps maintain data accuracy.
5. Allows authorised users to access information.

3. What is a DBMS?

DBMS stands for Database Management System. It is software that helps users create, store, organise, retrieve, modify and manage data in a database.

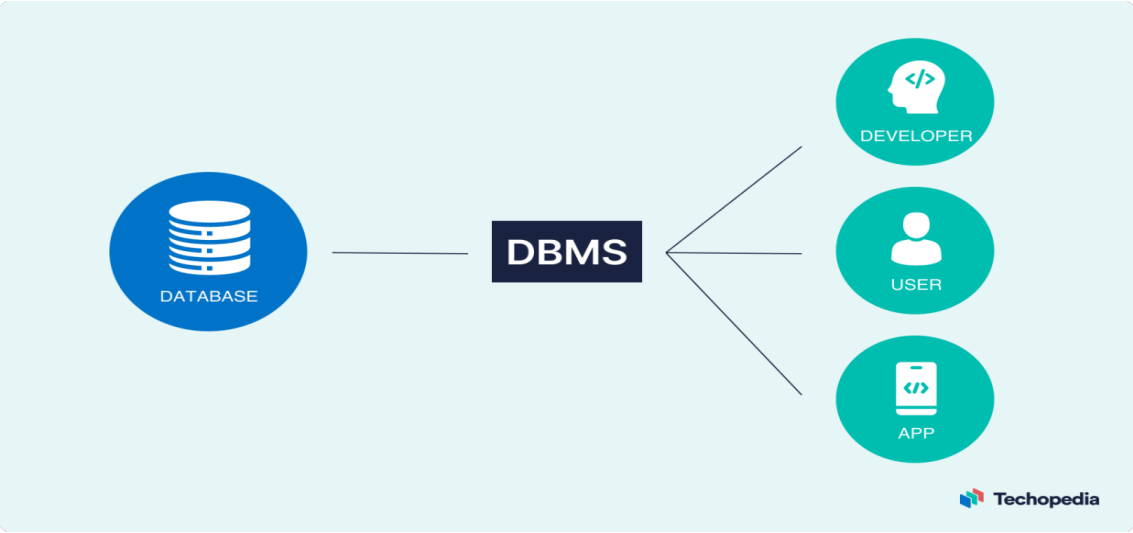
A DBMS acts as a connection between the user and the database. It makes database operations easier, faster and more secure.

In simple words, a database is a collection of data, whereas a DBMS is the software used to manage that data.

Examples of DBMS:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite
- Microsoft Access

How Does a DBMS Work?



Working:

1. The user sends a request to the DBMS.
2. The DBMS processes the request.
3. It communicates with the database.
4. The required data is retrieved or modified.
5. The DBMS provides the result to the user.

Main Functions of DBMS:

- Data storage and retrieval.
- Data updating and deletion.
- Data security and access control.
- Backup and recovery.
- Data sharing among multiple users.

4. Why Do We Need a DBMS?

In today's digital world, organisations generate and manage huge amounts of data every day. Managing such data manually or through separate files can become difficult, time-consuming and error-prone.

A DBMS provides an efficient way to store, organise and manage data.

Importance of DBMS:

1. Reduces Data Redundancy: Avoids unnecessary repetition of the same data.
2. Improves Data Security: Protects information from unauthorised access.
3. Easy Data Access: Helps users search and retrieve information quickly.
4. Maintains Data Consistency: Helps ensure that updated information remains accurate across the database.
5. Data Sharing: Allows multiple authorised users to access data.
6. Backup and Recovery: Helps restore data in case of accidental loss or system failure.

5. File System vs DBMS

A file system is a traditional method of storing data in separate files and folders. A DBMS is specialised software that manages data in an organised database.

Both methods are used to store information, but they differ in their features, efficiency and management.

Comparison between File System and DBMS

Basis	File System	DBMS
Data Storage	Separate files	Organised database
Data Redundancy	More chances of duplication	Reduces duplication
Security	Basic file security	Advanced access control
Data Sharing	Difficult to manage	Easier sharing
Backup	Often needs manual handling	Backup and recovery facilities
Data Searching	Can be time-consuming	Faster and easier
Data Consistency	Difficult to maintain	Better consistency control
Multi-user Access	Limited coordination	Controlled multi-user access

Advantages of DBMS over File System:

- Better organisation of data.
- Improved security.
- Easy data retrieval.
- Reduced data duplication.
- Better backup and recovery facilities.

6. Main Components of a DBMS

A DBMS consists of several components that work together to store, process and manage data efficiently.

The main components are:

1. Hardware

Hardware refers to the physical devices required to operate a database system, such as computers, servers, hard disks and storage devices.

2. Software

Software includes the DBMS application, operating system and other programs used to access and manage the database.

3. Data

Data is the most important component of a DBMS. It includes the actual information stored in the database, such as names, addresses, marks and records.

4. Procedures

Procedures are the instructions and rules used to operate, maintain, secure and manage the database.

5. Database Access Language

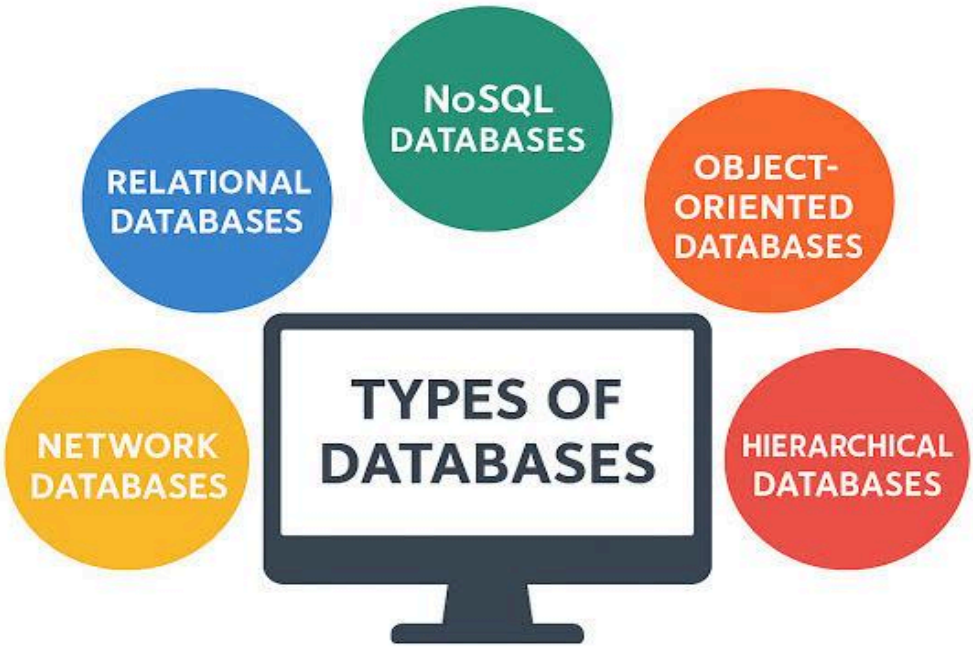
A database access language is used to communicate with the database. SQL (Structured Query Language) is one of the most commonly used database languages.

6. People / Users

People are the individuals who interact with the database system. They include database administrators, database designers, developers and end users.

7. Types of Database Models

A database model defines how data is organised, stored and connected within a database. Different database models are used depending on the requirements of an organisation.



8. SQL: The Language of Databases

SQL stands for Structured Query Language. It is a standard language used to communicate with relational database management systems.

SQL allows users to create tables, insert data, retrieve information, update existing records and delete unwanted data.

Main Categories of SQL Commands

Category	Full Form	Purpose
DDL	Data Definition Language	Defines database structure
DML	Data Manipulation Language	Inserts and modifies data
DQL	Data Query Language	Retrieves data
DCL	Data Control Language	Controls access and permissions
TCL	Transaction Control Language	Manages transactions

Common SQL Commands

Command	Use
CREATE	Creates a database or table
INSERT	Adds new records
SELECT	Retrieves records

UPDATE	Modifies existing records
DELETE	Removes records
ALTER	Changes table structure
DROP	Deletes a database object

9. Advantages and Limitations of DBMS

A DBMS provides many benefits in managing information. However, like other technologies, it also has some limitations.

Advantages of DBMS

1. Data Security: Protects information from unauthorised users.
2. Data Integrity: Maintains the accuracy and correctness of data.
3. Reduced Redundancy: Minimises unnecessary duplication of data.
4. Easy Data Access: Makes searching and retrieving information easier.
5. Data Sharing: Allows multiple authorised users to access information.
6. Backup and Recovery: Helps recover information after system failures.
7. Data Consistency: Helps maintain uniform and updated information.

Limitations of DBMS

1. High Cost: Some DBMS software and hardware can be expensive.
2. Complexity: Requires technical knowledge to manage.
3. Maintenance: Needs regular updates and maintenance.
4. Training: Users and administrators may require proper training.
5. System Failure: A database failure can affect many users.
6. Resource Usage: Large databases may require significant memory and processing power

10.Conclusion

A Database Management System (DBMS) is an important technology used to store, organise, retrieve and manage data efficiently. It helps individuals and organisations handle large amounts of information while maintaining data security, accuracy and consistency.

DBMS is widely used in banking, education, healthcare, railway reservations, e-commerce and many other fields. Understanding the basic concepts of databases, database models, keys and SQL provides a strong foundation for learning advanced database management concepts.

In today's digital world, DBMS plays an essential role in making information management easier, faster and more reliable.